



ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ



Distribute System

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TP 5

Networking and Internetworking

1. What is a gateway? Describe in detail

A gateway is a network device or software system that connects two or more networks that may use different communication protocols, architectures, or data formats. Its primary role is to enable communication between networks that would otherwise be incompatible.

A gateway acts as the entry point and exit point of a network. When data is sent from one network to another, the gateway receives the data, interprets it, performs any necessary protocol conversion, and forwards it to the destination network. Because of this, gateways often operate at multiple layers of the OSI model, especially the network layer and above.

In practical networks, a gateway is commonly implemented as a router. For example, in a home or office network, the router serves as the default gateway that connects local devices to the Internet. In addition to forwarding packets, gateways often perform functions such as Network Address Translation (NAT), firewall filtering, traffic management, and security enforcement. Without gateways, communication between different networks would not be possible.

2. Explain the main 5 cryptographic techniques in detail

Cryptography is the science of securing information by transforming it into a form that unauthorized users cannot understand. The five main cryptographic techniques are described below.

1) Symmetric Encryption

Symmetric encryption is a technique where the same secret key is used for both encrypting and decrypting data. The sender encrypts the original message using the secret key, and the receiver uses the same key to decrypt the message back to its original form.

This technique is very fast and efficient, making it suitable for encrypting large amounts of data. However, its main weakness is key distribution. If the secret key is intercepted during transmission, the attacker can decrypt the data. Because of this limitation, symmetric encryption is often combined with other cryptographic techniques.

Examples of symmetric encryption algorithms include AES and DES. Symmetric encryption is commonly used in VPNs, disk encryption, and secure file storage.

2) Asymmetric Encryption

Asymmetric encryption uses two different but mathematically related keys: a public key and a private key. The public key is shared openly and is used to encrypt data, while the private key is kept secret and is used to decrypt the data.

This approach solves the key distribution problem of symmetric encryption, since the public key can be shared safely. However, asymmetric encryption is slower and requires more computational power. Because of this, it is often used only to exchange keys or authenticate users, rather than encrypt large data sets.

Common asymmetric algorithms include RSA and Elliptic Curve Cryptography (ECC). Asymmetric encryption is widely used in secure web communication (HTTPS), SSH, and digital certificates.

3) Hashing

Hashing is a cryptographic process that converts input data of any size into a fixed-length output called a hash value or digest. Hashing is a one-way process, meaning it is computationally infeasible to reverse the hash to obtain the original data.

A good hashing algorithm ensures that even a small change in the input data produces a completely different hash value. This property is known as the avalanche effect. Hashing is not used for encryption or decryption, but rather for data integrity verification.

Hashing is commonly used to store passwords securely and to verify that data has not been altered during transmission. Popular hashing algorithms include SHA-256 and SHA-1.

4) Digital Signatures

Digital signatures are used to verify the authenticity, integrity, and origin of a message or document. The process begins by creating a hash of the original message. This hash is then encrypted using the sender's private key, forming the digital signature.

The receiver decrypts the digital signature using the sender's public key and compares the resulting hash with a newly computed hash of the received message. If the hashes match, the message is verified as authentic and unchanged.

Digital signatures provide authentication, integrity, and non-repudiation, meaning the sender cannot deny sending the message. They are widely used in secure emails, online transactions, and software distribution.

5) Message Authentication Code (MAC / HMAC)

A Message Authentication Code is a cryptographic technique used to ensure that a message has not been altered and that it comes from a legitimate sender. It combines a secret key with the message and a hashing algorithm to produce a code.

The receiver recalculates the MAC using the same secret key and compares it with the received MAC. If they match, the message is considered authentic and intact. MACs are commonly used in network protocols and secure communications.

3. Explain the differences between public and private IP addresses in detail

An IP address is a unique numerical identifier assigned to devices in a network. Public and private IP addresses serve different purposes in networking.

Public IP addresses are globally unique and are used to identify devices on the Internet. They are assigned by Internet Service Providers and allow devices to communicate across different networks worldwide. Every public IP address is unique to avoid conflicts on the Internet.

Private IP addresses are used within local networks such as homes, offices, and schools. These addresses are not directly accessible from the Internet and can be reused across multiple private networks. Private IP addresses improve security and conserve the limited number of public IP addresses.

Private IP ranges include 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, and 192.168.0.0 to 192.168.255.255.

4. What are the differences between Telnet and SSH? Describe in detail

Telnet and SSH are network protocols used to remotely access and manage computers. Telnet is one of the earliest remote access protocols, but it transmits all data, including usernames and

passwords, in plain text. This makes Telnet highly vulnerable to eavesdropping and security attacks.

SSH, which stands for Secure Shell, was developed to replace Telnet. SSH encrypts all data transmitted between the client and the server, ensuring confidentiality and integrity. SSH also supports secure authentication methods such as password authentication and public key authentication.

Telnet uses port 23, while SSH uses port 22. Due to its strong security features, SSH is widely used in modern network administration.

5. Explain VPNs in detail

A Virtual Private Network (VPN) is a technology that allows users to create a secure connection to a private network over the public Internet. VPNs encrypt all transmitted data, preventing unauthorized users from intercepting or modifying the communication.

VPNs are commonly used by organizations to allow remote employees to securely access internal resources. They are also used by individuals to protect privacy and hide their IP addresses. VPNs operate by creating an encrypted tunnel between the user's device and the VPN server.

There are two main types of VPNs: remote-access VPNs, which allow individual users to connect to a network, and site-to-site VPNs, which connect entire networks together securely.

6. Internet routing scalability problem and solution

The Internet consists of millions of networks and devices, making it impossible for any single router to store routing information for all destinations. To address this problem, the Internet uses a hierarchical routing scheme.

Networks are divided into Autonomous Systems (AS), each controlled by an organization such as an ISP. Within an AS, routers use Interior Gateway Protocols like OSPF or RIP. Communication between different ASes is handled by the Border Gateway Protocol (BGP). This approach reduces routing complexity and allows the Internet to scale efficiently.

7. Task of an Ethernet switch in detail

An Ethernet switch is a networking device that connects multiple devices within a local area network. It receives data frames and forwards them only to the intended destination device using MAC addresses.

The switch maintains a MAC address table that maps MAC addresses to specific ports. By forwarding data selectively rather than broadcasting it to all devices, switches reduce network congestion and collisions. Ethernet switches operate at the Data Link Layer of the OSI model and significantly improve network performance and efficiency.

8. How IP address uniqueness is ensured on the Internet

IP address uniqueness is ensured through a centralized global management system. The Internet Assigned Numbers Authority (IANA) manages the global IP address pool. IANA allocates address blocks to Regional Internet Registries, which then distribute them to ISPs.

ISPs assign IP addresses to customers in a controlled manner. This hierarchical allocation process ensures that no two devices on the Internet use the same public IP address simultaneously.

9. Best technique to calculate message transmission time

The best technique to calculate message transmission time is delay analysis. This technique considers all factors that contribute to the time taken for data to travel from sender to receiver.

The total transmission time includes transmission delay, propagation delay, processing delay, and queuing delay. By calculating each component and summing them, network engineers can accurately estimate message transmission time and evaluate network performance.

10. Security threats on the Internet in detail

Threats to processes include malicious software such as viruses, worms, spyware, and Trojan horses. These threats can damage systems, steal sensitive data, and disrupt normal operations.

Threats to communication channels include eavesdropping, packet sniffing, and man-in-the-middle attacks. These attacks compromise data confidentiality and integrity during transmission.

Denial of Service attacks aim to make systems unavailable by overwhelming them with excessive traffic. Distributed Denial of Service attacks involve multiple sources and can cause severe service disruption.